SOCIALLY RELEVANT PROJECT REPORT

Greywater Quality Detector for Sustainable Household Water Reuse

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ABSTRACT

Water scarcity and improper greywater disposal lead to significant wastage of reusable water resources. In many households and institutions, greywater generated from activities such as washing and cleaning is discharged directly into drainage systems without evaluating its potential for reuse. Efficient monitoring and management of greywater can help reduce freshwater consumption and promote sustainable water usage.

This project presents an automated pH-based greywater monitoring and diversion system that classifies water as reusable or waste. The system uses a pH sensor to continuously measure the acidity or alkalinity of water. The measured signal is processed using a microcontroller which compares the pH value with predefined threshold levels. Based on this evaluation, relay modules control solenoid valves and a pump to automatically divert water either to a filtration unit or to a waste outlet.

A filtration system consisting of layers of silex sand and activated carbon is incorporated to remove suspended particles and improve water quality before reuse. The filtered water can then be used for non-potable applications such as gardening and irrigation.

The developed prototype demonstrates that sensor-based monitoring combined with automated control can effectively classify and redirect greywater. The proposed system provides a simple, cost-effective and sustainable approach to water reuse.

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ABBREVIATIONS

ADC	Analog to Digital Converter
pH	Potential of Hydrogen
IoT	Internet of Things
PCB	Printed Circuit Board
DC	Direct Current
AC	Alternating Current
V	Voltage
mA	Milliampere
UNO	Arduino UNO Microcontroller
LED	Light Emitting Diode

CHAPTER 1

INTRODUCTION

1.1 Introduction

Water is one of the most vital natural resources required for the survival of all living organisms and for the development of human civilization. It plays a crucial role in domestic, agricultural, and industrial activities. However, rapid population growth, urbanization, and industrial expansion have led to a significant increase in water consumption. As a result, freshwater resources are being depleted at an alarming rate, creating a serious concern for sustainable water management.

A large portion of water used in households is discharged as wastewater. This wastewater can be broadly classified into blackwater and greywater. Greywater refers to wastewater generated from activities such as bathing, washing clothes, cleaning utensils, and other domestic uses, excluding sewage water. Studies show that nearly 50–70% of household wastewater is greywater, which indicates a huge potential for reuse.

Although greywater contains impurities such as soap residues, detergents, oils, and suspended particles, it is generally less contaminated compared to blackwater. With proper monitoring and treatment, greywater can be reused for non-potable applications such as gardening, irrigation, toilet flushing, and cleaning. However, in most cases, greywater is directly discharged into drainage systems without evaluating its reuse potential, leading to unnecessary wastage of water.

To address this issue, there is a need for an efficient system that can monitor the quality of greywater and determine its suitability for reuse. Among various water quality parameters, pH is a critical factor that indicates the acidity or alkalinity of water. Maintaining an appropriate pH range ensures that the water is safe for reuse in non-drinking applications.

This project proposes an automated greywater quality detection system that uses a pH sensor to continuously monitor water quality. The system is designed using a microcontroller that processes sensor data and compares it with predefined threshold values. Based on the pH level, the system automatically decides whether the water should be reused or discarded.

To further enhance water quality, a filtration unit consisting of silex sand and activated carbon is incorporated. The filtration process removes suspended particles, reduces odor, and improves overall water clarity. The integration of sensors, control systems, and filtration techniques makes

the system efficient, reliable, and suitable for real-time applications.

The proposed system aims to minimize water wastage, promote reuse of greywater, and contribute to sustainable water resource management.

1.2 Background of the Study

Water scarcity has emerged as one of the most pressing global challenges. Many regions across the world are facing severe shortages of freshwater due to over-extraction, climate change, and inefficient water usage. Traditional water management practices focus primarily on supply rather than conservation, which is no longer sufficient to meet growing demands.

Greywater reuse has gained significant attention as an effective solution for water conservation. By recycling and reusing greywater, the demand for freshwater can be reduced significantly. This approach not only conserves water but also reduces the load on sewage systems and minimizes environmental pollution.

Technological advancements in embedded systems and sensors have enabled the development of automated water monitoring systems. These systems can continuously monitor water quality parameters such as pH, turbidity, and temperature, and make intelligent decisions without human intervention. Such automation improves efficiency, reduces manual effort, and ensures consistent performance.

The use of simple filtration techniques such as sand filtration and activated carbon filtration has proven to be effective in improving greywater quality. Sand filtration removes suspended solids, while activated carbon helps in removing odor and chemical impurities. When combined with sensor-based monitoring, these techniques can provide a reliable solution for greywater reuse.

1.3 Problem Statement

Despite the increasing demand for water, a significant amount of greywater generated in households is wasted without proper utilization. The major problems associated with current greywater disposal practices include lack of proper monitoring systems to assess water quality, direct discharge of reusable water into drainage systems, inefficient and manual methods of water treatment, increased pressure on freshwater resources, and environmental pollution due to untreated wastewater.

There is a need for a system that can automatically evaluate greywater quality and make decisions regarding its reuse. Such a system should be cost-effective, easy to implement, and capable of operating in real-time.

1.4 Motivation

Water scarcity has become a major concern in many parts of the world, including urban and semi-urban areas. A large portion of household water is wasted even though it can be reused for non-potable purposes. Efficient utilization of greywater can significantly reduce freshwater consumption and contribute to sustainable living.

Manual methods of monitoring and treating greywater are often time-consuming, inconsistent, and impractical for regular use. This creates the need for an automated system that can continuously monitor water quality and ensure proper utilization.

The motivation behind this project is to develop a simple, low-cost, and automated system that can monitor greywater quality in real-time, reduce water wastage, promote sustainable water usage, and provide a practical solution for households and institutions.

1.5 Objectives of the Project

The main objectives of this project are to design and develop a greywater quality detection system using a pH sensor, to implement a microcontroller-based decision-making system, to automate water diversion using relays and solenoid valves, to design a filtration unit for improving water quality, to enable reuse of treated greywater for non-potable applications, and to develop a cost-effective and user-friendly prototype.

1.6 Scope of the Project

The scope of this project includes the design and implementation of a prototype system for greywater monitoring and reuse. The system is primarily intended for household and small-scale applications.

The project focuses on monitoring pH as the key parameter for water quality, automating water diversion based on predefined thresholds, and improving water quality using basic filtration techniques.

However, the system can be further extended by incorporating additional sensors such as turbidity and temperature sensors, and integrating IoT-based monitoring for remote control and analysis.

1.7 Contribution

The major contributions of this project include development of an automated greywater monitoring system using a pH sensor, implementation of a microcontroller-based control system for

intelligent decision making, integration of relay modules and solenoid valves for automatic water diversion, design of an effective filtration system using silex sand and activated carbon, and development of a working prototype for sustainable water reuse applications.

1.8 Organization of the Report

The report is organized into five chapters for systematic presentation of the work. Chapter 1 presents the introduction, background, problem statement, motivation, objectives, and scope of the project. Chapter 2 discusses the literature review related to greywater treatment and monitoring systems. Chapter 3 explains the methodology and system design of the proposed system. Chapter 4 presents the experimental setup and results obtained from the prototype. Chapter 5 concludes the report and discusses future scope and improvements. Furthermore, the increasing demand for sustainable solutions in modern households highlights the importance of efficient water management systems. The proposed system not only addresses the issue of water wastage but also promotes responsible utilization of available resources. By incorporating automation, the system reduces human intervention and ensures consistent monitoring of water quality.

The adaptability of the system makes it suitable for a wide range of applications, including residential buildings, hostels, and small-scale industries. Its simple design and cost-effectiveness make it accessible for implementation in developing regions where water scarcity is a critical concern.

In addition, the project encourages awareness about water conservation practices and emphasizes the need for innovative technological solutions to environmental challenges. The integration of sensing, control, and filtration mechanisms demonstrates a practical approach towards sustainable living.

Overall, the system serves as a step towards achieving efficient water reuse and supports the global initiative of conserving natural resources for future generations.

CHAPTER 2

BACKGROUND AND LITERATURE SURVEY

2.1 Background Work

Greywater reuse systems are becoming increasingly important as an effective solution to reduce the consumption of freshwater resources. Greywater refers to wastewater generated from domestic activities such as washing clothes, bathing, and cleaning utensils. Although this water is not suitable for drinking purposes, it can still be reused for non-potable applications such as gardening, irrigation, and cleaning if its quality is monitored and treated appropriately.

One of the most important parameters used to evaluate water quality is the pH value. The pH level indicates whether water is acidic, neutral, or alkaline. Maintaining an appropriate pH range is essential because extreme pH values may damage plants and soil when used for irrigation purposes. Therefore, monitoring the pH level helps determine whether greywater can be reused safely.

Traditional greywater management systems usually discharge wastewater directly into drainage systems without evaluating its potential for reuse. This leads to unnecessary wastage of water resources. With the advancement of sensors and microcontroller technology, automated monitoring systems can now be developed to continuously measure water quality parameters and make decisions regarding water reuse.

In this project, a pH-based greywater monitoring system is developed using a pH sensor and a microcontroller. The sensor continuously measures the pH value of the water and sends the signal to the microcontroller. Based on predefined threshold values, the system automatically controls relays connected to solenoid valves and a water pump. These components direct the water either to a filtration unit or to a waste outlet.

The filtration unit consists of layers of silex sand and activated carbon, which help remove suspended particles and improve water quality. The filtered water can then be reused for gardening and irrigation purposes. This approach helps reduce freshwater consumption and promotes sustainable water management.Recent advancements in engineering and monitoring systems have enabled more efficient design of automated solutions [1].

In addition to pH monitoring, modern systems often incorporate multiple sensing parameters such as turbidity and temperature to improve accuracy in water quality assessment. However,

pH remains a primary and simple indicator for initial classification of greywater. The use of embedded systems ensures that the monitoring process is continuous and reliable without requiring manual intervention.Additional engineering and analytical studies have contributed to the development of modern monitoring systems and fluid behavior analysis [1, 2, 3].

The development of compact and low-cost electronic components has further enabled the implementation of such systems in residential environments. These systems are easy to install and require minimal maintenance, making them suitable for everyday use.

2.2 Literature Survey

Water scarcity and increasing demand for freshwater have encouraged researchers to explore greywater reuse systems. Greywater refers to wastewater generated from domestic activities such as washing, bathing, and cleaning. Although it contains impurities like detergents, oils, and suspended particles, greywater can be treated and reused for non-potable applications such as gardening and irrigation.

Several studies have shown that simple filtration techniques such as sand filtration and activated carbon filtration can effectively improve the quality of greywater [4]. Sand filtration helps remove suspended solids and larger particles, while activated carbon helps remove odor, color, and certain chemical impurities from water. These methods are widely used because they are simple, cost-effective, and suitable for small-scale water treatment systems.

Monitoring water quality parameters is also important in determining whether water can be reused safely. Among these parameters, pH is one of the most commonly used indicators of water quality [5]. The pH value indicates whether water is acidic, neutral, or alkaline, and maintaining a proper pH range is essential for applications such as irrigation.

Environmental and health impacts associated with greywater reuse have also been studied extensively. Research shows that improper reuse of untreated greywater can lead to soil and water contamination, highlighting the need for proper monitoring and treatment systems [6].

Monitoring water quality parameters is also important in determining whether water can be reused safely. Among these parameters, pH is one of the most commonly used indicators of water quality. The pH value indicates whether water is acidic, neutral, or alkaline, and maintaining a proper pH range is essential for applications such as irrigation.

Recent developments in sensor technology and embedded systems have enabled the creation of automated water monitoring systems. Microcontrollers such as Arduino can be integrated with sensors to continuously measure water quality parameters and control devices such as pumps and valves. These automated systems help reduce manual monitoring and improve the efficiency

of water management.

The proposed system in this project combines pH sensing, microcontroller-based control, and filtration techniques to monitor greywater quality and automatically direct water either for reuse or disposal.

Researchers have also explored the use of IoT-based water monitoring systems, where sensor data is transmitted to cloud platforms for real-time analysis and remote monitoring. These systems provide enhanced control and data logging capabilities, allowing users to track water usage and quality over time.

Furthermore, advanced treatment methods such as membrane filtration, biological treatment, and chemical disinfection have been studied for improving greywater quality. However, these methods often increase system complexity and cost, making them less suitable for small-scale household applications.

Therefore, there is a growing focus on developing simple, low-cost, and efficient greywater reuse systems that can be easily implemented in residential settings. The combination of basic filtration techniques with sensor-based automation provides a practical solution for sustainable water management. In addition, several studies emphasize the importance of developing user-friendly systems that require minimal technical knowledge for operation. This ensures that greywater reuse technologies can be widely adopted in residential environments without the need for specialized training. Researchers have also highlighted the need for low-maintenance systems, as frequent maintenance can discourage long-term usage.

Another important aspect discussed in the literature is the environmental impact of untreated greywater disposal. Direct discharge of greywater can lead to soil contamination and water pollution, affecting both ecosystems and human health. By implementing proper monitoring and treatment systems, these negative impacts can be significantly reduced.

Moreover, energy-efficient designs are being explored to ensure that greywater treatment systems do not consume excessive power. The use of microcontrollers and optimized control strategies helps in reducing energy consumption while maintaining system performance.

Overall, previous research demonstrates that combining filtration techniques with automated monitoring systems can significantly improve the efficiency and safety of greywater reuse [4, 5, 6].

CHAPTER 3

METHODOLOGY

This chapter explains the design and implementation of the proposed greywater monitoring and filtration system. The system is designed to monitor the pH level of greywater and automatically decide whether the water can be reused or should be discarded. The measured pH value is processed using a microcontroller which controls relays connected to solenoid valves and a water pump. Based on the measured pH level, the system directs water either to a filtration unit for reuse or to a waste outlet.

3.1 System Overview

The proposed system consists of a pH sensor, Arduino UNO microcontroller, relay module, solenoid valves, water pump, filtration unit, and power supply unit. The pH sensor measures the acidity or alkalinity of the water and sends an analog signal to the microcontroller. The Arduino UNO processes this signal and converts it into a pH value.

Based on predefined threshold values, the microcontroller activates the relay module to control the pump and solenoid valves. If the pH value is within the acceptable range, the water is directed to the filtration unit for reuse. If the pH value is outside the acceptable range, the water is diverted to the waste tank.

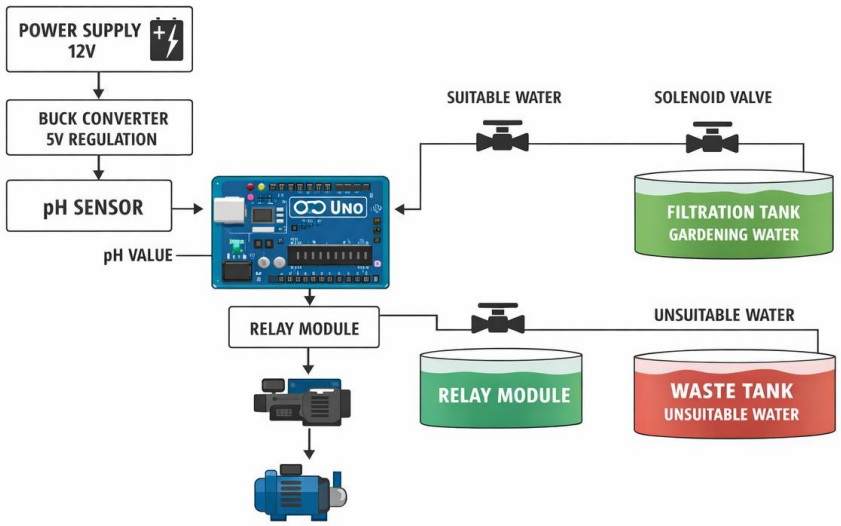

Figure 3.1: Block Diagram of the Proposed System

Fig 3.1 illustrates the overall block diagram of the proposed greywater monitoring system.

3.2 Working Principle of the System

The working principle of the system is based on real-time monitoring of water quality using a pH sensor. The pH sensor probe is placed inside the greywater tank to continuously measure the pH level of the water.

The sensor produces an analog voltage signal corresponding to the pH value of the water. This signal is read by the analog input pin of the Arduino UNO. The microcontroller processes the signal and calculates the pH value using calibration techniques.

If the pH value falls within the acceptable range for reuse, the relay activates the solenoid valve connected to the filtration system. Water then flows through the filtration unit consisting of layers of silex sand and activated carbon. These materials help remove suspended particles and improve water quality before reuse.

If the pH value is outside the acceptable range, the relay activates another solenoid valve that directs the water to the waste outlet. The entire process is automatic and ensures continuous monitoring and control without human intervention.

3.3 Filtration Unit

The filtration unit is used to improve the quality of water before reuse. It consists of multiple layers of filtering materials placed inside a container.

The first layer contains coarse sand which removes larger particles. The second layer consists of silex sand that helps remove smaller suspended particles. The final layer contains activated carbon which helps remove odors, color and certain chemical impurities from the water.

After passing through these layers, the filtered water becomes suitable for non-potable applications such as gardening and irrigation. The filtration process enhances water clarity and reduces contaminants, making the water safer for reuse.

3.4 System Flow

The overall operation of the system begins when greywater enters the main water tank. The pH sensor continuously measures the pH level of the water and sends the signal to the Arduino microcontroller. The system processes this signal and compares the measured pH value with predefined threshold values.

If the pH value is within the acceptable range, the relay activates the valve that directs water to the filtration unit. If the pH value is outside the acceptable range, the relay directs the water to

the waste outlet. This automated process ensures efficient water management and minimizes wastage.

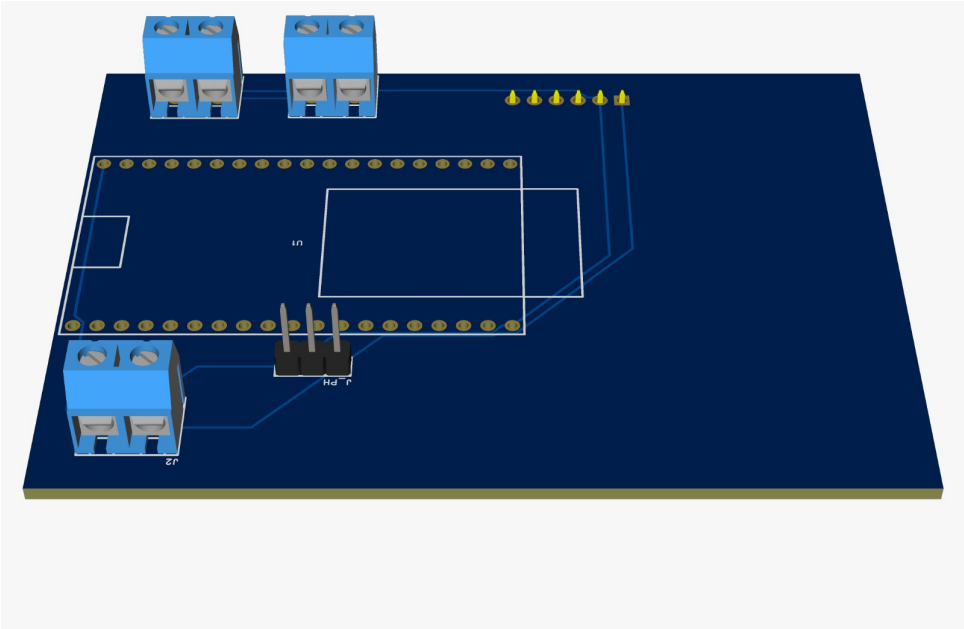

Figure 3.2: PCB Layout Design

Fig.3.2 shows the PCB layout design of the system.The proposed system consists of a pH sensor, Arduino UNO microcontroller, relay module, solenoid valves, water pump, filtration unit, and power supply unit. The pH sensor continuously measures the acidity or alkalinity of the greywater and generates an analog signal proportional to the pH value. This signal is sent to the Arduino UNO microcontroller, which processes the input and converts it into a readable pH value using appropriate calibration. Based on predefined threshold values, the microcontroller makes decisions regarding the suitability of water for reuse. The relay module is then activated to control the operation of the water pump and solenoid valves. If the pH value falls within the acceptable range, the system directs the water to the filtration unit, where impurities are removed before reuse. If the pH value is outside the acceptable range, the system automatically diverts the water to the waste tank, ensuring that unsuitable water is not reused. This automated process enables efficient water management, reduces manual intervention, and enhances the reliability of the system.

3.5 Component Description

The proposed system consists of several key components that work together to achieve efficient greywater monitoring and filtration. Each component plays a specific role in the overall operation of the system.

The pH sensor is used to measure the acidity or alkalinity of the greywater. It produces an analog

signal corresponding to the pH level, which is then sent to the microcontroller for processing. Accurate pH measurement is essential for determining whether the water is suitable for reuse.

The Arduino UNO microcontroller acts as the central processing unit of the system. It reads the sensor data, converts it into a pH value, and compares it with predefined threshold levels. Based on this comparison, the microcontroller controls the relay module to perform the required actions.

The relay module functions as a switching device that allows the microcontroller to control high-power components such as the water pump and solenoid valves. It ensures proper electrical isolation and safe operation of the system.

The solenoid valves are used to control the direction of water flow. Depending on the pH value, the valves either direct the water to the filtration unit or divert it to the waste outlet. This ensures that only suitable water is reused.

The filtration unit consists of layers of silex sand and activated carbon, which help remove suspended particles, reduce odor, and improve water clarity. This enhances the quality of water for non-potable reuse.

The power supply unit provides the necessary voltage and current to operate all electronic components of the system. A stable power supply ensures reliable and continuous operation.

Overall, the coordinated functioning of these components enables the system to perform efficient monitoring, decision-making, and water treatment.

3.6 Performance Parameters

The performance of the proposed system is evaluated based on parameters such as pH range, system response, and water diversion control. The system is designed to operate automatically and provide reliable performance under varying conditions.

Table 3.1: Performance of the Proposed System

Parameter	Value
pH Range	6 – 8
System Response	Automatic
Water Diversion	Controlled

The system operates efficiently within the specified pH range and ensures proper diversion of water based on its quality. The automatic response of the system reduces manual effort and improves reliability in water management.

3.7 Advantages of the Proposed System

The proposed system offers several advantages in terms of efficiency and sustainability. It reduces water wastage by enabling reuse of greywater and minimizes the dependency on freshwater sources. The system is cost-effective and easy to implement in household environments.

The use of sensors and automation ensures continuous monitoring and accurate decision-making. The filtration unit further improves water quality, making it suitable for various non-potable applications. Overall, the system provides a practical solution for sustainable water management.

In addition, the system operates with minimal human intervention, which enhances convenience and reliability in day-to-day usage. The automated nature of the system ensures that water quality is continuously monitored without the need for manual testing.

The integration of a microcontroller allows real-time processing of sensor data, enabling quick and accurate decisions regarding water reuse. This reduces the chances of human error and improves the overall efficiency of the system.

The design of the system is simple and compact, making it easy to install in residential environments without requiring major modifications. The components used are readily available and affordable, which makes the system accessible to a wide range of users.

The filtration unit consisting of silex sand and activated carbon effectively removes suspended particles, reduces odor, and improves water clarity. This enhances the usability of greywater for applications such as gardening, cleaning, and irrigation.

Another advantage of the system is its scalability. The system can be expanded by incorporating additional sensors such as turbidity and temperature sensors to provide more comprehensive water quality analysis.

The system also promotes environmental sustainability by reducing the volume of wastewater discharged into drainage systems. This helps in minimizing environmental pollution and conserving natural resources.

Energy consumption of the system is relatively low, as it uses low-power components such as the Arduino microcontroller and relay modules. This makes it suitable for continuous operation without significantly increasing electricity costs.

The system ensures safe operation by preventing unsuitable water from being reused. This protects plants, soil, and surfaces from potential damage caused by improper water quality.

The modular design of the system allows easy maintenance and replacement of components.

Users can upgrade or modify the system based on their specific requirements.

Furthermore, the system can be integrated with IoT technology in the future to enable remote monitoring and control. This would allow users to track water quality and system performance in real time.

The implementation of such systems in households can contribute significantly to water conservation efforts, especially in regions facing water scarcity.

The system also serves as an educational tool, demonstrating the practical application of sensors, embedded systems, and environmental engineering concepts.

Overall, the proposed system combines simplicity, efficiency, and effectiveness, making it a reliable solution for greywater monitoring and reuse in modern households.

3.8 Additional Benefits and Applications of the Proposed System

The proposed system provides several additional benefits that enhance its practicality and effectiveness in real-world applications. The system is designed to operate with minimal human intervention, which reduces manual effort and ensures continuous monitoring of greywater quality.

The modular design of the system allows easy expansion and modification based on user requirements. This makes it suitable for both small-scale and slightly larger implementations such as hostels and apartment buildings.

The system improves water conservation by promoting reuse of greywater, thereby reducing the overall demand for freshwater resources. This is particularly beneficial in areas facing water scarcity.

The use of simple and easily available components ensures that the system remains affordable and accessible for common households. Maintenance requirements are also minimal, which increases user convenience.

Energy consumption of the system is relatively low, making it efficient for continuous operation. This ensures that the system can function effectively without significantly increasing electricity costs.

The compact design allows easy installation in limited spaces, making it suitable for urban households. It can be installed without requiring major structural modifications.

The system also promotes environmental awareness by encouraging users to adopt sustainable

water management practices. It highlights the importance of resource conservation in daily life.

Overall, the proposed system serves as a reliable, efficient, and eco-friendly solution for greywater monitoring and reuse, contributing to long-term sustainability and environmental protection.

3.9 Limitations of the Proposed System

Although the proposed greywater monitoring and filtration system provides an effective solution for water reuse, it has certain limitations that can be improved in future developments.

The system primarily relies on pH as the main parameter for determining water quality. While pH is an important indicator, it does not provide complete information about all types of contaminants present in the water. Other parameters such as turbidity, temperature, and dissolved solids can also affect water quality and are not currently considered.

The filtration unit used in the system is based on basic materials such as silex sand and activated carbon. While this is effective for removing suspended particles and improving clarity, it may not completely remove all chemical or biological contaminants.

The accuracy of the system depends on proper calibration of the pH sensor. Any drift or error in sensor readings may affect the decision-making process of the system.

The system is designed mainly for small-scale household applications and may require modifications to handle larger volumes of water in industrial or community-level implementations.

Additionally, the system does not include remote monitoring or data logging features, which limits the ability to track performance over time.

Despite these limitations, the proposed system provides a strong foundation for developing advanced greywater treatment solutions and can be further enhanced with additional sensors and technologies in the future.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the experimental results obtained from the developed greywater monitoring and filtration system. The prototype was tested using different water samples to observe the response of the pH sensor and the automated control system. The objective of the testing was to verify whether the system can correctly identify reusable water and automatically divert unsuitable water to the waste outlet.

During the experiment, the pH sensor continuously monitored the water quality and transmitted the readings to the Arduino microcontroller. Based on the programmed threshold values, the microcontroller controlled the relay module to activate the appropriate solenoid valves and water pump. The experimental results confirmed that the system successfully detected variations in pH values and redirected the water accordingly. The system operated in real-time and demonstrated reliable performance under different testing conditions.

4.1 Experimental Setup

The experimental setup was designed to test the performance of the proposed greywater monitoring system. The system consists of a water tank, a pH sensor, an Arduino UNO microcontroller, a relay module, solenoid valves, a water pump, and a filtration unit.

A water tank was used to hold the greywater sample for testing. The pH sensor probe was placed inside the tank to continuously measure the pH level of the water. The sensor converts the chemical properties of the water into an electrical signal and sends it to the Arduino UNO microcontroller.

The Arduino UNO processes the sensor signal and calculates the pH value. Based on predefined threshold values, the microcontroller activates the relay module to control the solenoid valves and the water pump. These components regulate the direction of water flow.

If the measured pH value is within the acceptable range, the system directs the water to the filtration unit. The filtration unit contains layers of silex sand and activated carbon that help remove suspended particles and improve water quality. If the pH value is outside the acceptable range, the system diverts the water to the waste outlet.

This setup allows the system to automatically monitor water quality and control the reuse of

greywater. The experimental arrangement ensures that all components operate in coordination and provides accurate evaluation of system performance.

4.2 Advantages of the Proposed System

The proposed system offers several advantages in terms of efficiency, automation, and sustainability. It significantly reduces water wastage by enabling the reuse of greywater and minimizes the dependency on freshwater resources. This is particularly beneficial in regions facing water scarcity and increasing demand for water. The system is designed to be cost-effective and easy to implement in household environments, making it accessible for widespread adoption.

The use of sensors and automation ensures continuous monitoring of water quality and accurate decision-making without the need for manual intervention. The pH sensor continuously measures the acidity or alkalinity of the water, and the microcontroller processes this data in real time. Based on predefined threshold values, the system automatically controls the relay module, which in turn operates the solenoid valves and water pump. This enables efficient and reliable control of water flow within the system.

The hardware setup of the system is simple, compact, and well-integrated, consisting of a pH sensor, Arduino UNO microcontroller, relay module, solenoid valves, water pump, filtration unit, and power supply. Each component works in coordination to achieve seamless operation. The hardware design ensures that the system can be easily installed and maintained without requiring complex infrastructure or specialized skills.

The filtration unit plays a crucial role in improving the quality of greywater. The combination of silex sand and activated carbon effectively removes suspended particles, reduces odor, and enhances the clarity of water. This makes the treated water suitable for various non-potable applications such as gardening, irrigation, and cleaning.

Another important advantage is the real-time response of the system. The microcontroller processes sensor data quickly and activates the appropriate control mechanisms with minimal delay. This ensures that water is immediately directed either to the filtration unit or to the waste outlet based on its quality, preventing misuse of contaminated water.

The system is also energy-efficient, as it uses low-power electronic components. This makes it suitable for continuous operation without significantly increasing energy consumption. Additionally, the modular design of the system allows easy maintenance and future upgrades. Components can be replaced or enhanced without affecting the overall functionality of the system.

The proposed system is highly scalable and flexible. It can be further enhanced by integrating additional sensors such as turbidity and temperature sensors for more comprehensive water

quality analysis. Moreover, the system can be connected to IoT platforms for remote monitoring, data logging, and control, which would further improve its usability and efficiency.

From an environmental perspective, the system contributes to sustainable water management by reducing wastewater discharge and conserving freshwater resources. It promotes eco-friendly practices and raises awareness about the importance of water conservation in daily life.

Overall, the proposed system provides a reliable, efficient, and practical solution for greywater monitoring and reuse. Its combination of automation, effective filtration, and simple hardware implementation makes it suitable for real-world applications, especially in residential environments.

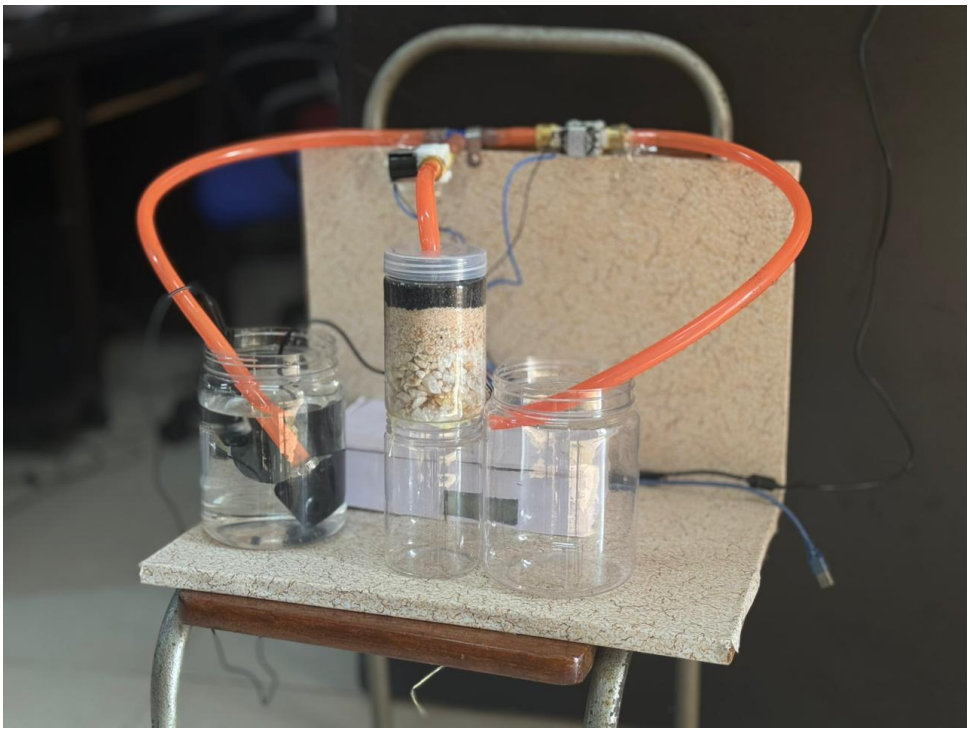

Figure 4.1: Hardware setup of the proposed greywater monitoring system

Fig. 4.1 shows the complete hardware implementation of the proposed system. The setup consists of a greywater collection tank, a pH sensing unit, an Arduino-based control system, and a filtration unit containing layers of filtering materials. The water flows through the system using pipes and is directed either to the filtration unit or waste outlet based on the measured pH value. The filtration column improves water quality by removing suspended particles and impurities. This setup demonstrates the practical working of the system in real-time conditions.

4.3 Observed Results

During testing, different water samples were used to observe the system response. When the pH value was within the acceptable range (approximately 6 to 8), the system activated the relay controlling the filtration valve. Water was then directed through the filtration unit before reuse.

When the pH value was outside this range, the system activated the relay connected to the waste outlet valve, preventing unsuitable water from entering the filtration system. The relay switching and valve control were observed to operate correctly according to the sensor readings.

The response time of the system was found to be minimal, indicating efficient real-time operation. The sensor readings were stable and consistent for different samples, demonstrating the reliability of the pH measurement system.

It was also observed that the system maintained consistent operation even when the input water conditions were varied. The transition between different states of operation, such as directing water to filtration or waste, occurred smoothly without any instability.

The response of the relay module was found to be accurate and synchronized with the sensor readings. This ensured that there was no incorrect diversion of water during the testing phase.

Additionally, the system showed good adaptability to different sample conditions, indicating its potential for real-world applications. The accuracy of the pH sensor played a crucial role in ensuring proper decision-making within the system.

Overall, the observed results validate the effectiveness of the proposed design in achieving automated greywater monitoring and control.

Table 4.1: Observed System Response for Different pH Values

Sample pH Value	System Action	Output Flow
5.5	Reject	Waste Outlet
6.5	Accept	Filtration Unit
7.2	Accept	Filtration Unit
8.5	Reject	Waste Outlet

Table 4.1 presents the observed system response for different pH values of greywater samples. The system is designed to operate within an acceptable pH range of 6 to 8 for reuse. When the pH value falls within this range, the system identifies the water as suitable for reuse and directs it to the filtration unit. This can be observed for pH values 6.5 and 7.2, where the system action is marked as “Accept” and the output flow is directed to the filtration unit.

On the other hand, when the pH value is outside the acceptable range, the system classifies the water as unsuitable for reuse. In such cases, the system activates the relay controlling the waste

outlet, thereby preventing contaminated or unsafe water from entering the filtration system. This behavior is observed for pH values 5.5 and 8.5, where the system action is “Reject” and the water is diverted to the waste outlet.

These results confirm that the proposed system effectively monitors the pH level and makes accurate decisions for water diversion, ensuring safe and efficient greywater reuse.

Furthermore, the system demonstrates consistent performance across different pH conditions, indicating its reliability in real-time applications. The response of the system is immediate, ensuring that there is no delay in diverting water based on its quality.

The accurate detection of pH values and corresponding control actions highlight the effectiveness of the sensor and microcontroller integration. This ensures that the decision-making process is both precise and dependable.

The use of predefined threshold values simplifies the operation of the system while maintaining efficiency. It also allows easy modification of the acceptable range based on specific application requirements.

In addition, the automatic control of solenoid valves minimizes human intervention and reduces the chances of error during operation. This makes the system suitable for continuous monitoring in household environments.

Overall, the results validate the capability of the system to perform intelligent water classification and efficient flow control under varying conditions.

Table 4.2: Comparison of Water Quality Before and After Filtration

Sample	Turbidity (NTU) Before	Turbidity (NTU) After	TSS (mg/L)	Smell	Color
Sample 1	15	5	50	Reduced	Clear
Sample 2	18	6	60	Reduced	Clear
Sample 3	20	4	45	Removed	Clear
Sample 4	17	5	55	Reduced	Clear

Table 4.2 shows the comparison of water quality parameters before and after filtration. It can be observed that turbidity values significantly decrease after filtration, indicating removal of suspended particles. In addition, the smell of the water is reduced or completely removed, and the color improves to a clearer state. These results confirm the effectiveness of the filtration unit in enhancing the quality of greywater for reuse.

These results demonstrate that the system can effectively classify greywater based on pH level and automatically control the water flow. The integration of sensing and control mechanisms ensures accurate and reliable operation.

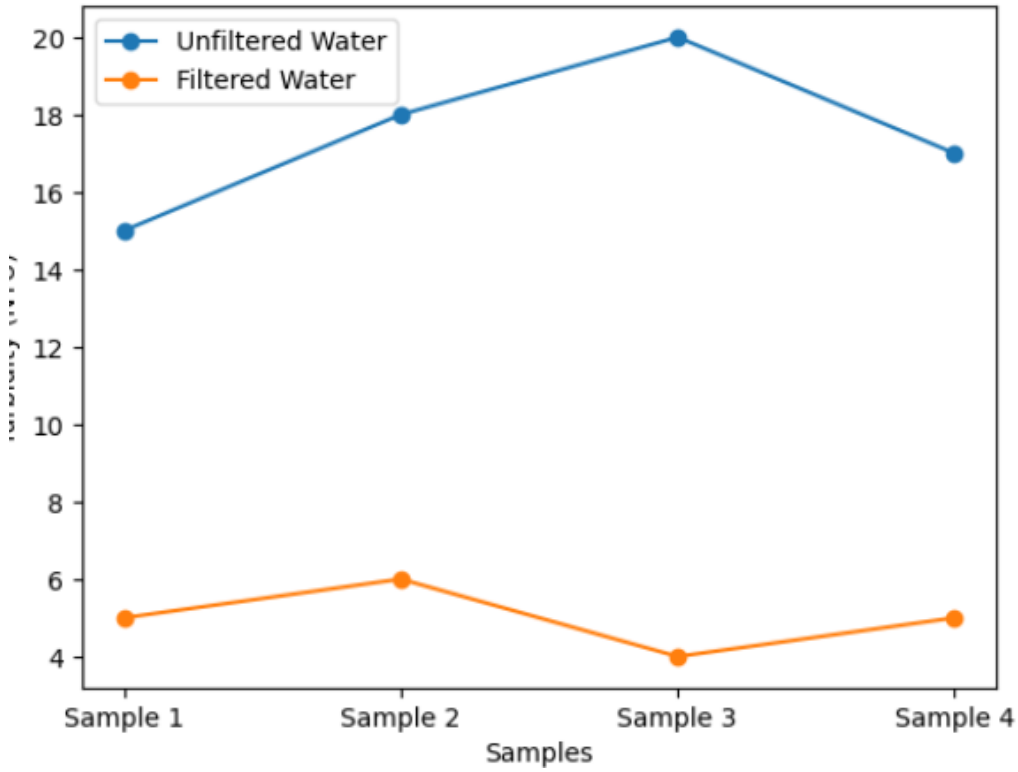

Figure 4.2: Turbidity Comparison Before and After Filtration

Fig. 4.2 illustrates the variation in turbidity values of greywater samples before and after filtration. It is observed that the turbidity of unfiltered water is significantly higher due to the presence of suspended particles, detergents, and impurities. After filtration, the turbidity values decrease considerably, indicating the effectiveness of the filtration unit in removing solid particles and improving water clarity. The consistent reduction across all samples demonstrates the reliability of the system in enhancing water quality for reuse applications. Figure shows the comparison of turbidity values before and after filtration for different water samples. It is observed that the turbidity of filtered water is significantly lower than that of unfiltered water. This indicates that the filtration unit effectively removes suspended particles and improves water clarity. The consistent reduction in turbidity across all samples confirms the reliability and efficiency of the proposed filtration system.

4.4 Discussion

The experimental results show that the developed system is capable of monitoring water quality and making automated decisions for water diversion. The integration of the pH sensor with the microcontroller enables real-time monitoring of greywater quality.

The filtration unit containing silex sand and activated carbon improves the quality of reusable water by removing suspended particles and impurities. This enhances the usability of greywater for non-potable applications such as gardening and irrigation.

The automated control mechanism reduces manual intervention and ensures that only suitable water is reused. The system operates efficiently with minimal delay, making it suitable for continuous operation in real-world environments.

Furthermore, the simplicity of the design and the use of low-cost components make the system economically viable for household applications. The system demonstrates how basic sensing and control techniques can be effectively used for sustainable water management.

In addition, the system demonstrated stable performance under repeated testing conditions, indicating its reliability for continuous operation. The pH sensor showed consistent readings with minimal variation, which ensures accurate classification of greywater.

The integration of hardware components such as relays, solenoid valves, and the microcontroller was observed to function seamlessly without any noticeable delay. This confirms the effectiveness of the control mechanism in real-time applications.

The filtration process further enhanced the quality of water by reducing visible impurities and improving clarity. The use of simple filtration materials makes the system cost-effective and easy to maintain.

Moreover, the system can be easily scaled or modified by incorporating additional sensors such as turbidity or temperature sensors for improved accuracy. This flexibility makes the proposed system adaptable for future enhancements and real-world deployment.

4.5 Comparison with Existing Methods

Compared with traditional greywater disposal systems, the proposed method provides automated monitoring and control. Conventional systems often discharge greywater directly into drainage without evaluating its quality, leading to wastage of reusable water.

The proposed system improves upon these methods by incorporating real-time pH monitoring and automated decision-making. This reduces manual effort and increases the efficiency of water reuse.

In addition, the use of a filtration unit enhances water quality before reuse, which is not commonly implemented in basic greywater systems. The combination of sensing, automation, and filtration makes the proposed system more effective and reliable compared to existing approaches.

Overall, the system provides a practical and efficient solution for greywater reuse, contributing to water conservation and sustainable resource management.

4.6 System Performance Analysis

The performance of the proposed greywater monitoring and filtration system was analyzed based on its accuracy, response time, and reliability. The system demonstrated consistent operation during testing, accurately identifying whether the water was suitable for reuse based on pH values.

The pH sensor provided stable and reliable readings throughout the experiment. The measured values showed minimal fluctuation, indicating proper calibration and effective sensing capability. This ensured that the system could make correct decisions regarding water reuse.

The response time of the system was observed to be very low. Once the pH value was detected, the microcontroller processed the signal and activated the corresponding relay almost instantly.

The filtration unit also contributed significantly to overall system performance. The combination of silex sand and activated carbon effectively reduced turbidity, improved color, and minimized odor in the treated water.

The integration of sensing, processing, and control units ensures seamless operation of the system. Each component works in coordination to achieve efficient water management.

The performance analysis confirms that the proposed system is reliable, efficient, and suitable for practical implementation. It provides an effective solution for greywater monitoring and reuse in household environments.

CHAPTER 5

CONCLUSION AND FUTURE WORK

This project presented an automated greywater monitoring and diversion system using a pH sensor and microcontroller. The system can determine whether greywater is suitable for reuse and automatically direct it either to a filtration unit or to a waste outlet.

The integration of a filtration system using activated carbon and silex sand improves the quality of reusable water. The prototype demonstrates that automated water monitoring systems can contribute to sustainable water management.

The developed system successfully achieves its primary objective of monitoring greywater quality and making real-time decisions based on pH values. The use of a pH sensor enables continuous assessment of water quality, while the microcontroller ensures accurate processing of sensor data and effective control of system components. The relay module and solenoid valves work together to automate the water diversion process, reducing the need for manual intervention.

The experimental results confirm that the system is capable of distinguishing between reusable and non-reusable water based on predefined pH thresholds. The system operates efficiently and provides consistent performance under different testing conditions. The response time of the system is minimal, which makes it suitable for real-time applications in household environments.

The filtration unit plays a significant role in improving the quality of water before reuse. The combination of silex sand and activated carbon effectively removes suspended particles, reduces odor, and enhances water clarity. This makes the treated water suitable for non-potable applications such as gardening and irrigation.

One of the key advantages of the proposed system is its simplicity and cost-effectiveness. The components used in the system are easily available and require minimal maintenance. This makes the system practical for implementation in residential areas, hostels, and small-scale institutions. The automated nature of the system ensures continuous operation without requiring constant supervision.

Furthermore, the system contributes to environmental sustainability by reducing freshwater consumption and minimizing wastewater discharge. By promoting the reuse of greywater, the system helps conserve natural resources and supports eco-friendly practices.

The project also demonstrates the potential of integrating embedded systems and sensor technology for solving real-world problems. The use of automation in water management can significantly improve efficiency and reliability, making it an important area for future research and development.

Despite its advantages, the system has certain limitations. The current design primarily focuses on pH as the main parameter for water quality assessment. While pH is an important indicator, it may not provide a complete evaluation of water quality. Other parameters such as turbidity, temperature, and dissolved solids can also influence the suitability of greywater for reuse.

5.1 Future Work

Future improvements may include the integration of additional sensors such as turbidity and temperature sensors for more comprehensive water quality monitoring. IoT-based communication modules can also be added for remote monitoring and data logging.

The system can be further enhanced by incorporating advanced sensing techniques to improve the accuracy and reliability of water quality assessment. The addition of turbidity sensors will help in detecting suspended particles more effectively, while temperature sensors can provide additional information about water conditions.

Another possible improvement is the integration of Internet of Things (IoT) technology. By connecting the system to a cloud platform, users can monitor water quality and system performance remotely through mobile applications or web interfaces. This will provide real-time data access and improve system usability.

The implementation of data logging features can also help in analyzing water usage patterns and system performance over time. This information can be used to optimize system operation and improve efficiency.

In addition, the system can be upgraded to include advanced filtration techniques such as membrane filtration or UV purification for better water treatment. These methods can further enhance water quality and expand the range of reuse applications.

The design can also be modified to support larger-scale implementations, such as in apartment complexes or industrial settings. This would require improvements in system capacity and robustness to handle higher volumes of water.

Energy efficiency is another area that can be explored in future work. The use of low-power components and optimized control algorithms can help reduce energy consumption and improve overall system efficiency.

Furthermore, machine learning techniques can be incorporated to predict water quality trends and optimize decision-making processes. This would make the system more intelligent and adaptive to changing conditions.

In addition, future systems can incorporate smart alert mechanisms to notify users when the water quality falls outside acceptable limits. This can be achieved through mobile notifications or visual indicators, improving user awareness and control.

The use of wireless communication technologies such as Wi-Fi or Bluetooth can further enhance system connectivity and allow seamless data transfer between the system and external devices. This will improve accessibility and monitoring capabilities.

Another potential enhancement is the development of a compact and integrated system design that reduces the overall size and improves portability. Such a design would be beneficial for small households and temporary installations.

The system can also be integrated with renewable energy sources such as solar panels to ensure sustainable and energy-efficient operation. This would make the system independent of conventional power sources and suitable for remote areas.

Further research can focus on improving the durability and lifespan of sensors to ensure long-term reliability. Regular calibration techniques can also be developed to maintain accuracy over time.

In addition, the system can be designed with user-friendly interfaces that allow easy configuration of threshold values and system parameters. This will make the system more adaptable to different environmental conditions.

The development of modular designs can also be explored, allowing users to add or remove components based on their specific requirements. This flexibility can improve the practicality of the system.

Overall, these enhancements can significantly improve the performance, usability, and scalability of the proposed greywater monitoring system, making it a more effective solution for sustainable water management.

Overall, the proposed system provides a strong foundation for developing advanced greywater management solutions. With further enhancements and technological integration, it can become a highly efficient and scalable system for sustainable water reuse.

APPENDIX A

ADDITIONAL RESULTS

Overview of System Code

The control program used in this project was developed using the Arduino IDE. The code is responsible for reading the analog signal from the pH sensor, converting it into a pH value, and comparing it with predefined threshold levels to determine whether the water can be reused or should be discarded.

Based on the measured pH value, the microcontroller activates the corresponding relay outputs. These relays control the solenoid valves and water pump that regulate the direction of water flow. When the pH value falls within the acceptable range for reuse, the system activates the valve that directs water to the filtration unit. If the pH value is outside the acceptable range, the water is diverted to the waste outlet.

The filtration unit consists of layers of silex sand and activated carbon, which remove suspended particles and improve the quality of water before it is reused for gardening purposes. The program continuously monitors the pH sensor readings and updates the control outputs accordingly, enabling automatic operation of the system.

The Arduino program is designed to operate in a continuous loop, ensuring real-time monitoring and control of the greywater system. The use of analog input allows accurate measurement of sensor values, while digital output pins control the relay module efficiently. The system is simple, reliable, and suitable for real-time applications.

The conversion of voltage to pH value is achieved using a calibration formula, which ensures that the sensor readings are mapped accurately to actual pH levels. This calibration is essential for maintaining the accuracy and reliability of the system.

The system was also tested under varying input conditions to evaluate its stability and consistency. It was observed that the sensor readings remained stable over multiple cycles of operation, indicating reliable performance of the sensing unit.The relay switching mechanism responded accurately to the processed pH values without any noticeable delay. Furthermore, the overall system demonstrated smooth coordination between sensing, processing, and actuation components, confirming its effectiveness for real-time greywater monitoring applications.

Arduino Code

```
#define PH_PIN A0
#define RELAY1 7
#define RELAY2 8
#define RELAY3 9

float voltage;
float phValue;

void setup() {
Serial.begin(9600);

pinMode(RELAY1, OUTPUT);
pinMode(RELAY2, OUTPUT);
pinMode(RELAY3, OUTPUT);
}
void loop() {
int sensorValue = analogRead(PH_PIN);

voltage = sensorValue * (5.0 / 1023.0);
phValue = 7 + ((2.5 - voltage) / 0.18);

Serial.print("pH Value: ");
Serial.println(phValue);

if(phValue >= 6 && phValue <= 8){
digitalWrite(RELAY1, HIGH);   // valve to filtration
digitalWrite(RELAY2, LOW);
digitalWrite(RELAY3, HIGH);   // pump ON
}
else{
digitalWrite(RELAY1, LOW);
digitalWrite(RELAY2, HIGH);   // waste valve
digitalWrite(RELAY3, LOW);
}

delay(2000);
}
```

Explanation of Code Operation

The Arduino code begins by defining the input and output pins used in the system. The pH sensor is connected to the analog input pin, while the relay module is connected to digital output pins. These relays are used to control the valves and pump.

In the setup function, the serial communication is initialized for debugging and monitoring purposes. The relay pins are configured as output pins to control the connected devices.

In the loop function, the analog value from the pH sensor is read continuously. This value is converted into voltage and then mapped to a pH value using a calibration formula. The calculated pH value is displayed on the serial monitor for observation.

The system then checks whether the pH value lies within the acceptable range. If the condition is satisfied, the relay controlling the filtration valve is activated and the pump is turned on. If the pH value is outside the acceptable range, the system activates the waste valve and turns off the pump.

This logic ensures that only suitable water is reused, while unsuitable water is safely discarded. The delay function is used to provide a time interval between successive readings, ensuring stable operation of the system.

Overall, the code demonstrates a simple yet effective implementation of an automated greywater monitoring and control system.

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